

TyreMind

Clean tyre degradation curves from noisy practice data — with confidence intervals that have been measured, not claimed.

A lap time is a bill with no itemisation. You paid 95 seconds. How much was the tyre? We separate the tyre from fuel burn, track evolution and traffic, then state how sure we are and prove that number is honest.

4.3×

better than the closest published model at recovering a degradation rate

95.2%

measured interval coverage over 69,206 real laps, against 95% claimed

74%

of 77 real races where the standard method returns a physically impossible answer

13

of our own hypotheses tested and refuted, all published

What makes this different in one sentence

Every competitor can show a degradation number. We show one that is right, that says how confident it is and is actually right that often, that proves the standard method is broken in three races out of four, and that states mathematically how much of the answer is data and how much is assumption. Then we route each question to whichever of nine models the evidence says is best at it, including away from our own.

THE PROBLEM

Four things move a lap time at once. Only one of them is the tyre, and it is the only one you cannot see.

CAUSE	DIRECTION	ROUGHLY WORTH	OBSERVABLE?
Tyre wearing out	slower	0.03–0.08 s/lap	No — this is the target
Fuel burning off	faster	0.081 s/lap	Physics, pinned by prior
Track rubbering in	faster	~0.9 s per session	Shared across the field
Traffic / dirty air	slower	up to 1.3 s	We built an index for it

Why this is hard and not merely fiddly

The thing hiding the answer is the same size as the answer, and points the opposite way. Fuel is worth about as much as the tyre. Ignore it and you do not get a slightly wrong number — you get the wrong sign.

The brief, verbatim

"Create a predictive model that strips out external noise from practice sessions to generate clean tyre performance degradation curves. Includes post-race validation tools to compare predicted wear against actual race-day pace."

The honest core of the whole project

Public F1 data contains **no measured tyre wear**. Pirelli measures it, the FIA and teams see it, the public never does. We searched Pirelli media, FIA documents, Kaggle, Hugging Face, academic replication packages, OpenF1, Ergast and the timing schema itself. It does not exist.

So we cannot grade ourselves against truth on real data. Everything that follows is designed around that constraint instead of pretending it away.

In 74% of real races, the textbook method says the tyres got **faster** as they wore out.

Tyres do not regenerate. You do not need to know the true wear rate to know a negative one is wrong — which is why this is the most defensible single result we have.

74.0%

of 77 Grands Prix return a negative degradation rate on at least one compound

53.4%

of 208 individual compound stints, same failure

0.0725

naive rate error s/lap against known truth — 20× ours

0%

of its 95% intervals actually contained the answer

The falsification argument, which is the part judges have not seen before

Most validation needs ground truth. This one does not. A method that reports tyres improving with age has refuted itself, and we can count how often that happens across four seasons of real racing. **exp14 is a validation leg that survives the total absence of wear labels**, and no paper in our literature review runs it.

And it is not a strawman — this is what mentors challenged us on

A mentor said "regression will never work". They were right, and this proves it. But we also built the **fixed** version: a pooled regression with fuel, track and traffic terms. It is rung 3 of our own ladder, it beats us at forecasting lap times, and we route that question to it. Slide 09.

Real data, and synthetic data built to be hostile to us.

Real — downloaded and cleaned ourselves

Season corpus, 2022–2025	203 sessions · 91,867 laps
Telemetry, 27 features per lap	202 sessions · 147,550 laps
OpenF1 measured intervals	70 races
3D track geometry with elevation	6 circuits
NASA C-MAPSS turbofan benchmark	4 datasets
Research papers, 11 topic folders	31 PDFs

We invented a traffic index, then validated it against measurement

Derived from lap start times alone, no positional data needed. Checked against OpenF1's measured 4 Hz gaps: **median $p = -0.798$ across 66 races, correct sign in 66 of 66**, worst single race -0.512 , largest p -value 3×10^{-19} . Stint-level agreement $p = 0.912$.

Synthetic — and why there are two generators

Generator 1 is deliberately hostile: Student-t noise where our estimator assumes Gaussian, and 25% scrubbed sets so tyre age does not start at zero.

Generator 2 exists because a generator shaped like the estimator cannot falsify it. Lap time comes from a line integral over a speed profile, fuel acts through mass, traffic through downforce loss. **There is no rate parameter anywhere** — the truth is a counterfactual, differenced.

The caveat we always state first: we wrote both. Generator 2 removes the functional-form advantage. It removes nothing else.

Seven cleaning filters, every one logged

Every session ships a receipt. Monza 2024: the API gave 1,008 laps, we kept 921, and the file says exactly where the other 87 went. Outliers by median + $3 \times \text{MAD}$, not mean and standard deviation — a safety car inflates a standard deviation enough to hide itself.

Ten stages from a timing feed to an answer a strategist can act on.

01

Ingest

Four fields from the official timing feed. A parquet drops in and is served.

02

Clean

7 filters, each logged. A receipt per session.

03

Features

run_id, traffic_index, lap_in_run — all derived, none given.

04

Fit

Kalman forward, RTS smoother back. 6 parameters, ~12 s.

05

Calibrate

Conformal. Where "95%" is made to mean 95%.

06

State

One object, every field carrying its own standard deviation.

07

Oracle + Decision

Probabilities on one path, the pit optimiser and 18 veto rules on the other.

08

Verifier

Every sentence becomes typed claims, checked against the state before it ships.

09

Agent

Role-adapted language across four surfaces. Retrieval, not generation.

10

Router

Each question to the model measured best at it. Slide 09.

Stages 5, 8 and 10 are what no competitor has

Everything else on this diagram is standard engineering. The calibration layer makes the confidence honest, the verifier makes it impossible to publish a sentence the state does not support, and the router makes the product an orchestration rather than one model with opinions.

The agent talks to the verifier, not to the model

There is **no write path**. An agent cannot change a fit, a threshold or a published number. It can only read state and phrase it, and the verifier rejects any claim whose units or magnitude do not match.

Three things look identical in the data.

Naming them, and pinning each one, is the method.

```

y[d,t] =  $\alpha[r]$       run intercept: car, setup, start fuel
        -  $A \cdot g(t)$     track evolution, whole field
        +  $\theta[t]$         residual track wobble
        +  $s[d,t]$         ← THE TYRE. What we want. Invisible.
        -  $\phi \cdot L[d,t]$  fuel burn-off, L = laps this run
        +  $\gamma \cdot TI[d,t]$  traffic
        +  $\epsilon$           noise
  
```

Latent state, evolving in TYRE AGE not session lap:

```

s[d,a+1] = s[d,a] + rate[d,a] + noise
rate[d,a+1] = rate[d,a] + noise ← lets a cliff appear
  
```

We never tell the model what shape degradation has

Because rate is free to drift, a cliff needs no special machinery — it is rate accelerating. Everyone else forces a straight line. We measured **2,827 real stints**; only 55.9% are linear. **The field-standard model is the wrong shape for 44% of real stints.**

① Fuel vs tyre — perfectly confounded

$$y = c + \beta(a_0 + f) - \phi \cdot f = (c + \beta \cdot a_0) + (\beta - \phi) \cdot f$$

The slope you can see is $\beta - \phi$, not β . More data does not help; it is algebra. Pinned by a physical prior on ϕ whose own uncertainty widens every interval we publish.

② Track evolution vs a uniform rate shift

Pinned by making track evolution a *saturating* curve $A(1 - e^{-kL})$. Rubber deposition genuinely saturates; tyre wear does not. Physically correct *and* identifying.

③ Tyre age vs session lap — our best idea

Every car pits at a different time, so on lap 30 one car is on 3-lap-old tyres and another on 25-lap-old ones. **The grid staggers itself.** Watch one car and you cannot do this; watch twenty and it solves itself **for free, with no assumption.**

Cappello & Hoegh fit one driver in one race. They structurally cannot use this.

Nine models rebuilt from their own papers.

On recovering a degradation rate, we are 4.3× the closest published one.

MODEL	RATE ERROR S/LAP	95% INTERVAL COVERED
TyreMind state-space	0.0037	100%
Pooled regression	0.0062	83%
Heilmeyer et al. — field standard	0.0112	100%
Cappello & Hoegh 2025 — closest paper	0.0158	38%
Fuel-corrected regression	0.0212	54%
Naive regression	0.0725	0%

Where we lose, stated before anyone finds it

Lap-time forecasting: 4th of 9. A pooled regression wins at 0.4088 against our 0.6484, so we route that question to it. **Real pit stops:** a tie, 5.92 ± 0.52 laps against 5.98 ± 0.52 for the closest paper — intervals that overlap that far are the same result and we record it as a tie.

But the pooled regression that wins the lap-time table comes **last** on those same 274 pit stops at 11.12 laps, answering only half of them. **Forecasting a lap time and explaining why it moved are different jobs.**

It survives a truth we did not write

On the independent physics generator our margin is $+0.0086 \pm 0.0017$, which is **2.6× larger** than on our own generator. But we also report that "the degradation rate" is three different quantities on a non-linear generator, and a different model wins each.

And it survives 11 hostile worlds

32 seeds across 11 generator regimes: **8 wins, 1 tie, 2 losses**. Biggest margin on the collinear worst case, exactly where it should help most. Both losses are on the track-evolution axis and one was **not predicted** — that is our clearest architectural weakness and it is in the result file.

Four regimes, measured not assumed

linear 1,580 · recovery 582 · cliff 337 · warm-up 328
cliff arrives at 71.9% of stint life

Our model said 95%. We checked. It was right 75.5% of the time. So we fixed it, and then measured it again.

That is lying with mathematics, and it is what every competing system ships — because nobody checks.

WHAT WE CLAIMED	WHAT WE DELIVERED	SAMPLE
95% — model's own error bars	75.5%	94 comparisons
95% — after conformal calibration	94.7%	94 comparisons
95% — live, adaptive conformal	95.2%	69,206 laps
90% — pit window	90.8%	held-out real stops
80% — pit window	83.1%	held-out real stops

Calibrated is easy. Calibrated *and narrow* is the product.

Anyone can be right 95% of the time by widening the window until it covers everything. For the **same 90% guarantee**, the window each model needs:

TyreMind		13 laps
Cappello & Hoegh		17 laps
Pooled regression		20 laps
Naive		22 laps

The most damning number about the standard method

Naive regression claims **47.3%** confidence on its pit window and delivers **2.5%**. Ours claims 31.9% and delivers 24.0% — still overconfident by 7.9 points, and we say so. Best of six, not perfect.

How the probability is produced

Not a softmax dressed up as a probability. We simulate **1,200 races per candidate pit lap under common random numbers**, so draw i read across all candidates is one coherent race, and we count the winners. That is a frequency.

How much is data, and how much is assumption

6.0% data-driven. Cramér–Rao bound 0.0161 against our 0.0221 — we sit at 1.38× the theoretical optimum. **No paper in our review states this number**. Cappello & Hoegh have the identical structure and never quantify their prior's contribution.

The product is not a model. It is a router.

Nine models, five tasks, three different winners — and we route away from ourselves twice.

QUESTION	ANSWERED BY	MARGIN	EVIDENCE
Forecast a lap time	Pooled regression	0.4088 vs our 0.6484	exp19
Recover the degradation rate	TyreMind	0.0037 vs 0.0062	exp19
When to pit	TyreMind (tie, recorded as one)	5.92 vs 5.98	exp22
Practice → race	Cappello & Hoegh	-0.882 vs our -1.154	exp27
How confident are you	TyreMind	-7.9% vs -8.9%	exp30

Two rules stop this being a way to launder a weak result

Every route cites a result file. The code refuses to construct a route without evidence — a route with no evidence is a preference, and preferences are how a system ends up routing to whatever its author likes.

A margin inside the noise is a tie, and ties break on honesty. Three models sit within one standard error on pit timing. Claiming a win on 0.06 laps would be exactly the cherry-picking this project exists to avoid.

The result we route away from, and why it matters more than the win

On predicting Sunday from Friday, every model on the bench scores **negative skill** — worse than ignoring practice and quoting that driver's own average. So the product **declines the forecast** rather than dressing a negative-skill number as a prediction.

We found that by building a test designed to kill our own earlier, flattering number. It worked.

"We tested nine models on five tasks. Three different models won. So we route each question to the one the evidence says is best — including away from our own model twice. Competitors ship one model and claim it wins everything."

The build fails if a route goes stale

Tests re-read the result files behind every route. Re-run an experiment, change a winner, and the build breaks rather than leaving the table asserting fiction.

The same code, unmodified, runs on jet engines.

Strip the motorsport words away and this is not a tyre problem. It is: **something wears out while it works, you cannot measure it directly, and several other things move the only signal you can see.**

NASA C-MAPSS — the most-cited benchmark in predictive maintenance

Turbofan engines run to failure with **published remaining-life labels**, which is exactly what motorsport data does not have. We pointed the same estimator at it with no tyre-specific code.

Remaining-life error, RMSE	22.66 cycles
Mean absolute error	17.94 cycles
Engines scored	100 of 100
Predicted early — the safe direction	44%

Said plainly, because a judge will look it up

Published deep models reach **15.98**. We are **not state of the art on this benchmark and have never claimed to be**. The claim is *transfer*: unmodified tyre code on jet engines, in the competitive range, with no retuning.

What *is* competitive is the uncertainty. A 2025 paper presents 93.5–95.2% coverage as a new contribution; our conformal layer reaches 95.2% over 69,206 laps.

What transferred with no change

- ▶ The latent state model — a level and a rate that drift
- ▶ Pooling a baseline across a fleet of units
- ▶ The entire uncertainty machinery
- ▶ Remaining-life projection to a threshold

What changed: "laps" became "flight cycles", "lap time" became a health index from 12 sensors, fuel and traffic switched off.

A physics check that costs nothing to show

We recover **circuit rotation direction** — clockwise or anticlockwise — from per-corner tyre energy alone, correct on **7 of 8 circuits**. It uses no wear data. If the energy model were wrong, it would get this wrong.

Where this goes next

Aviation, wind turbines, rail braking, mining haul fleets. Any asset with a degrading component, no direct sensor, and a confounded observable.

Four fields, twelve seconds, no GPU, no internet.

There is no integration project.

Affordability — the barrier is close to zero

Fit time per session ~12 s on a laptop

Hardware No GPU

Four seasons of data on disk 2.6 MB

Internet required None — runs offline

Dependencies numpy, scipy, pandas

Data licensing cost Zero — public timing

The neural network we built as a baseline took 36 s and came last of nine. Capacity was never the bottleneck.

Scalability

- ▶ Per-session fits are **independent** — 200 sessions is 200 parallel jobs
- ▶ The live path is one Kalman update per lap: **milliseconds**
- ▶ A new circuit needs **no retraining**, because nothing is trained
- ▶ New regulations arrive, we fit them that day

Who needs this, in order

1 · Junior formulae and club racing. F2, F3, F4, endurance privateers. No tyre modelling department, same physics, same public timing. This is the wedge.

2 · F1 teams. They already hold strictly more data than we require, so we fit their feed by construction. Our value is the calibration layer and the identifiability bound, not the state model.

3 · Anyone with a wearing asset. The NASA result on slide 10 is the proof that the engine is not motorsport-specific.

Adaptive to any data — the precise claim

We need four fields: lap_time, compound, tyre_age, lap_number. Anything richer is a bonus, never a requirement.

Where it does not adapt, honestly: wet running is excluded and reported as excluded; kerb strikes are impact, not gradual wear, so we show them in the residual; traffic-heavy stints under-report by about a quarter.

Trustworthy enough to adopt

615 automated tests, including structural ones that fail the build if someone bypasses the repairing data loader. Every session ships a data-quality receipt.

Thirteen of our own ideas failed. We published every one of them.

HYPOTHESIS WE TESTED	OUTCOME
Energy clock beats lap clock	No difference
True C1–C5 compound beats the weekend label	6.4% worse
Circuit geometry transfers between tracks	No effect, $p = 0.22$
Temperature explains our bias	$p = -0.022$, $p = 0.835$
Matching stint depth fixes the bias	Worse, $p = 0.016$
Driver style is a usable feature	0.88% worse
Telemetry energy predicts degradation	Wrong sign
Sector times break the collinearity	Identify <i>where</i> , not <i>how much</i>
Practice predicts race	Negative skill for every model
Measured energy transfers between circuits	Harmful, $p < 0.001$
Kerb exposure predicts degradation	Null — and not measurable publicly
Heteroscedastic noise fixes the traffic defect	Intervals got narrower. Reverted
A 10-state 4-corner physics EKF beats our 2-state model	43% worse at transfer, 25× slower

We built the more advanced model on purpose, to try to beat ourselves

A ten-state continuous-discrete Extended Kalman Filter: four wear states, four temperatures, thermal coupling, skewed-t noise, driven by measured per-corner loads. **1,967 lines, 118 tests, pre-registered before a line was written.**

It converged on every one of 140 real stints. It still lost: **43% worse** at carrying a compound parameter to a new circuit, at $p = 2 \times 10^{-6}$, and 304 s per session against a 60 s budget. So it does not ship.

And we caught our own circular test

The first run of that experiment said the new model was **28% better**. It was scoring the model's own in-sample fit against an out-of-sample baseline. Fixing the leak moved the result by 71 points and flipped its sign.

"A team that publishes thirteen refuted hypotheses and the bugs it found in its own work is a team whose surviving claims can be believed. No competing write-up in our literature review reports a single negative result."